

Vigneshwara Chinnadurai

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Professional Summary

Senior AI Architect & Data Scientist with **8+ years** of experience designing enterprise-scale **Agentic AI**, **Generative AI**, and Machine Learning solutions. Pioneer in building **production-grade Autonomous AI Agent systems** using **AWS AgentCore SDK**, with expertise in **Hybrid RAG pipelines**, **LLM-powered self-critique loops**. Delivered **\$277M+** measurable business impact across Fortune 500 clients.

Experience

Senior AI Architect (Senior Consultant)

Deloitte Consulting USI

January 2022 – Present

Bangalore, India

Disney TVA Schema Intelligence Chatbot / AWS Bedrock, Claude 3.7 Sonnet, Agent Forge SDK

- Architected **production-grade conversational AI chatbot** using custom **S3VectorsAgentRuntime** extending AWS AgentCore SDK to handle S3 Vectors' unique `metadata.text` storage pattern that standard SDK couldn't parse.
- Implemented **intelligent token management** with configurable `max_rag_context_size` trimming that dynamically truncates retrieval chunks to prevent context overflow.
- Innovated **dual-path KB retrieval**: SDK's `extract_retrieval_chunks` for standard `content.text` + custom fallback for `metadata.text` with automatic deduplication.
- Created **follow-up query resolution system** with heuristic-based pronoun detection + URN extraction, rewriting vague queries for improved RAG retrieval.
- Optimized memory retrieval latency from **17+ seconds to under 5 seconds** through memory hook optimization and session stashing before RAG augmentation.
- Built **KB write-back pipeline** with deterministic vector keys enabling immediate queryability of agent-generated summaries without sync delays.

Disney PGT Document Parser Agent / AWS Lambda, Textract, Claude 3.7 Sonnet, SAM

- Architected **intelligent multi-format document parsing pipeline** for Disney's upfront spec sheet extraction, processing **Excel, PDF, Images, and Email (.msg)** with type-specific processors (pandas/openpyxl, Textract async/sync, extract-msg).
- Designed **serverless event-driven architecture**: S3 Upload → Lambda Router → File Processor → Textract/pandas → Smart Chunker → Claude (Bedrock) → DynamoDB, deployed via **AWS SAM/CloudFormation**.
- Implemented **smart chunking strategy** for large documents exceeding Claude's context limit with header context preservation and parallel multi-pass extraction with merged results.
- Engineered **composite Proposal ID generation** system linking disparate document types (specs, emails, PDFs) to unified proposal records using deterministic hashing when explicit IDs are absent.

Disney ESPN Game Summary Platform / AWS Nova Premier, Lambda, Real-time Video AI

- Developed **GenAI highlight extraction platform** contributing to **\$231.76M AWS business expansion**. Architected **7-step serverless Lambda pipeline**: MediaLive → Kinesis → MediaPackage → S3 → Nova Premier → DynamoDB → Highlights.
- Designed **multimodal prompt engineering** for video analysis with confidence scoring per field, play-type classification, and precise timestamp extraction. Built `clean_response_text()` handling markdown fences and partial JSON.
- Expanded architecture from **NFL to Cricket, Rugby, and Soccer leagues** with 90% event capture rate.

HP Enterprise - CXO Talk to Data Platform / Databricks Genie, Azure OpenAI GPT-4o-mini

- Built **AI-powered conversational analytics platform** replacing traditional BI dashboards for executive decision-making. First implementation of **Databricks Genie's NL-to-SQL** capabilities with user token forwarding.
- Architected **multi-table semantic layer** (2 Fact + 11 Dimension tables) with YAML-based semantic definitions. Implemented **hallucination reduction** through output validation against schema definitions.

Toyota Demand Forecasting Engine / LSTM, N-BEATS, Prophet, PySpark

- Engineered enterprise-scale forecasting system managing **1.2 million SKU combinations** across 15 distribution centers, reducing annual inventory costs by **\$46.1 Million** with 70% WMAPE.
- Implemented **hybrid LSTM architecture** with dual-layer Lambda expansions, **zero-entry "+1 trick"** for MinMax scaling, and **ensemble model selection** (LSTM/N-BEATS/SVR/Prophet) per SKU characteristics.
- Designed `orchestrator.py` with **multiprocessing pool** executing parallel forecasts with YAML-driven multi-environment configuration.

Halliburton Financial Predictive Modeling / N-BEATS, SARIMAX, Deep Learning

- Developed **deep learning forecasting solution** for Accounts Payable/Receivable prediction using **N-BEATS** and **SARIMAX** ensemble, achieving **98% accuracy** in invoice dispute prediction.
- Reduced financial forecast **delivery time by 98%** through automated pipeline optimization and multi-variate time series modeling capturing both trend and seasonality patterns.

Firm Initiatives: Agentic AI & MLOps Estimation Frameworks

- Created **industry-first estimation frameworks** for Agentic AI and MLOps project sizing with 24+ iterations, **adopted across multiple Deloitte practices** for consistent project scoping.

Senior Software Engineer (Machine Learning)

WIPRO Technologies

February 2018 – January 2022

Bangalore, India

WIPRO Holmes (R&D) / LSTM, Reg-exp, Deep Learning

- Enhanced Contract Review tool using **Bi-LSTM + CRF architecture** for sequence labeling of legal clauses in MSAs and SOWs. The **bidirectional LSTM** captured long-range contextual dependencies while the **CRF layer** learned valid BIO tag transitions (B-CLAUSE → I-CLAUSE), preventing invalid sequence predictions. Combined with **character-level embeddings** for OOV legal terminology and **domain-specific regex patterns** for structured entities (dates, parties, amounts), improved extraction accuracy from **76% to 93%**.

Huawei Video (R&D) / DIP, Vector DB, Deep Learning

- Created data model for **real-time recommendation pipelines** using **Deep Interest Network (DIN)** for user behavior modeling. Built **attention-weighted embeddings** capturing user-item interactions, stored in **Redis** for low-latency serving + **Milvus** for vector similarity retrieval. Integrated with **RASE (Huawei Recommendation-as-a-Service)** for 3 pipelines: **Hot Broadcast** (trending long-form), **You May Like** (personalized long-form), and **Waterfall E2** (short-video infinite scroll).

Shell Plc / CNN, Transfer Learning, Machine Learning

- Built **CNN with VGG19 Transfer Learning** in Keras to detect small symbols (valves, pumps, instruments) in P&ID drawings. Implemented **Hough Circle Transform + TensorFlow Object Detection API** for geometric shape recognition, automating 2D-to-3D schematic conversion.
- Developed **OCR pipeline** combining **Tesseract** (for structured text) + **AWS Textract** (for tables/forms) with custom image preprocessing (contrast enhancement, adaptive thresholding, noise reduction) for legacy engineering document digitization.

Personal R&D Projects

Autonomous Multi-Agent Strategic Analysis Engine | *Gemini 2.5 Flash, Hybrid RAG, Self-Refinement*

- **Genuinely Agentic Architecture:** LLM-powered planning where **AnalystAgent** autonomously decomposes complex prompts into executable multi-step plans without predefined workflows.
- **Hybrid Retrieval Pipeline:** Custom BM25 sparse + Gemini dense embeddings with **Reciprocal Rank Fusion (RRF)** + LLM-powered reranking (0-1 relevance scoring).
- **Self-Critique & Refinement Loop:** Rubric-based critique (Depth/Evidence/Causality/Actionability) with automatic NEEDS_REFINEMENT detection triggering re-retrieval cycles.
- **Dynamic Tool Generation:** Runtime code synthesis creating callable Python functions from LLM output. **Graph Memory** with semantic triple extraction for persistent knowledge.
- **Real data integration:** SEC EDGAR XBRL, Google News RSS, Gemini Search Grounding. **Cost Guard System** with budget caps and aggressive caching.

Quantum Portfolio Optimization Engine (MBA Project) | *D-Wave Quantum Annealing, QUBO, Simulated Annealing*

- **QUBO Formulation from First Principles:** Quadratic Unconstrained Binary Optimization with risk terms (covariance), return rewards (mean returns), and cardinality constraints via penalty functions.
- **Classical-Quantum Benchmark:** Side-by-side scipy max-Sharpe vs. quantum annealing with hyperparameter sweep over *risk & return*, penalty coefficients.
- **D-Wave Leap QPU Integration:** Production-ready execution with **EmbeddingComposite** for automatic qubit mapping. Efficient frontier visualization.

LeetCode AI Companion (Android) (Personal Development) | *Kotlin, Jetpack Compose, Gemini API, GraphQL*

- **Jetpack Compose UI** with Material Design 3. LeetCode GraphQL integration for daily challenges. **Gemini generation pipeline** with retry logic.
- **GitHub Push Automation:** One-tap export of solutions to repository. Calendar event integration for completion tracking. Mermaid flow viewer with gestures.

Education

Manipal University
Master of Business Administration, Analytics and Data Science
Anna University
Bachelor of Engineering, Electrical & Electronics

Jaipur, India
August 2024 – June 2026
Chennai, India
August 2014 – May 2017

Certifications

AWS Certified Machine Learning – Specialty | Amazon Web Services (Oct 2021)
Delivery Accreditation - AI Agents | ServiceNow
Configure AI Applications to Optimize Search Results | Google
Deep Learning Specialization | Coursera/Udacity

Technical Skills

Agentic AI & RAG: Agent Forge SDK (AWS AgentCore), Strands Agents, Multi-Agent Orchestration, LLM Planning & Critique, Self-Refinement Loops, Hybrid Retrieval (BM25 + Dense + RRF), LLM Reranking, S3 Vectors, Token-aware Trimming
LLMs & Foundation Models: Claude 3.7 Sonnet, GPT-4/4o-mini, AWS Nova Premier, Gemini 2.5 Flash, Llama 3, Amazon Titan Embed v2
Quantum Computing: D-Wave Quantum Annealing, QUBO Formulation, Simulated Annealing, D-Wave Leap, Binary Quadratic Models
Deep Learning: Transformers (BERT, DiT), CNN (VGG19, ResNet), LSTM, Bi-LSTM, Deep Interest Network (DIN), N-BEATS, Prophet
Cloud & MLOps: AWS (Bedrock, SageMaker, Lambda, S3, MediaLive, Kinesis, SQS, Identity Manager, Media Package, Media Convert), Azure (OpenAI), GCP (Vertex AI), Docker, Kubernetes, Airflow, DataBricks
Languages & Data: Python, PySpark, SQL, Kotlin (Android), Pandas, NumPy, TensorFlow, PyTorch

Key Achievements

\$46.1M annual inventory savings at Toyota | \$231.76M AWS expansion via Disney ESPN | 30% latency reduction (70s→50s) | 93% contract extraction accuracy | 1.2M SKUs forecasting scale | 4+ Fortune 500 production deployments